

**A Capstone Project Report on
Surface Water and Drinking Water Contaminants**

CS 896: Data Science Capstone

By

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Submission Date: May 4, 2026

Abstract:

Water quality is an important environmental and public health concern, especially in regions where communities depend on both surface water and groundwater for drinking purposes. This project focuses on analyzing contaminant patterns in drinking water systems across Kansas and examining how these patterns relate to surface water conditions.

Two datasets were used in this study: a municipal drinking water dataset and a surface water dataset obtained from the Water Quality Portal (WQP). The drinking water data required significant preprocessing, including unit standardization, handling of non-numeric values, and creation of derived variables such as standardized concentration (ppb), exceedance indicators, and outlier flags.

Exploratory analysis identified Barium, Fluoride, Selenium, Nitrate, and Chromium as the most frequently reported contaminants. Among these, nitrate showed significantly higher concentration levels compared to others, increasing from approximately 2800 ppb in 2019 to nearly 5000 ppb in 2024. Regression analysis indicated that year alone does not explain variation when all contaminants are combined, highlighting the importance of contaminant-specific analysis.

The surface water dataset enabled spatial analysis due to the availability of geographic coordinates. Monitoring locations were mapped, and nitrate was analyzed across locations and years. The results showed that nitrate concentrations vary across sites and fluctuate over time without a clear linear trend. Site-level analysis further confirmed that higher values are distributed across multiple locations rather than concentrated in a single area.

A key limitation of this study is the absence of geographic identifiers in the drinking water dataset, which prevents direct linkage between surface water sources and drinking water systems. Therefore, the findings are presented as two related analyses rather than a direct causal relationship.

Overall, this project demonstrates the importance of data preprocessing, exploratory analysis, and spatial context in understanding environmental datasets and provides a foundation for future work involving integrated water quality analysis.

Introduction

Water quality is a critical factor in protecting public health and maintaining environmental sustainability. In many regions, including Kansas, communities depend on both surface water sources such as rivers and reservoirs, and groundwater systems for their drinking water supply. Because of this, understanding how contaminants behave in these water systems is essential for ensuring safe and reliable water for communities.

Water contamination can occur due to multiple factors, including agricultural runoff, industrial activities, natural geological conditions, and urban development. These factors can introduce various chemical substances into water systems, some of which may exceed safe regulatory limits. Monitoring these contaminants and analyzing their patterns over time helps identify potential risks and supports better decision-making in water management.

This project was carried out as part of a data science capstone effort to analyze water quality data and explore possible relationships between surface water and drinking water contamination. The initial motivation was to understand whether contamination observed in surface water sources could influence the quality of drinking water supplied to communities.

To address this, the project uses two datasets: a municipal drinking water dataset and a surface water dataset obtained from the Water Quality Portal (WQP). The drinking water dataset provides information on contaminant levels reported by community water systems, while the surface water dataset includes measurements collected from monitoring locations across Kansas, along with geographic coordinates.

The project focuses on identifying key contaminants in drinking water, analyzing their concentration trends over time, and evaluating whether these patterns align with observations from surface water data. In particular, nitrate was identified as an important contaminant and analyzed in detail across both datasets.

Initially, the goal was to directly link surface water monitoring locations with drinking water systems to understand how contamination flows between the two. However, during the analysis, it was found that the drinking water dataset does not include geographic identifiers such as latitude and longitude. Because of this, a direct spatial linkage between the datasets was not possible.

As a result, the project was structured into two parts: a statistical analysis of drinking water contaminants and a spatial analysis of surface water data. While these analyses cannot be directly connected, they provide valuable insights into contaminant behavior within each system.

Overall, this project demonstrates how data analysis techniques can be applied to environmental datasets to identify patterns, understand variability, and highlight important limitations that affect real-world data science problems.

Problem Statement and Objectives

Understanding how surface water contamination relates to drinking water quality is an important problem in environmental data analysis. In many cases, surface water bodies serve as

primary sources for municipal water systems, and changes in these sources may influence the quality of treated drinking water. Identifying patterns in contaminant levels across these systems can provide useful insights for monitoring and management.

The main problem addressed in this project is to analyze contaminant levels in drinking water systems across Kansas and examine whether similar patterns can be observed in surface water data. The goal is to understand which contaminants are most important, how their concentrations change over time, and whether surface water data provides additional context for these patterns.

To address this problem, the project was designed with the following objectives:

- Identify the most frequently reported contaminants in the drinking water dataset and analyze their concentration levels.
- Examine trends in contaminant concentrations over time to understand whether values are increasing, decreasing, or remaining stable.
- Evaluate exceedance patterns by comparing observed values with regulatory limits (MCL) to identify potential risk areas.
- Analyze variability in contaminant data by identifying outliers and understanding extreme values.
- Perform spatial analysis of surface water data to understand how measurements vary across different monitoring locations.
- Focus on nitrate as a key contaminant and analyze its behavior in both drinking water and surface water datasets.
- Explore the possibility of linking surface water contamination with drinking water systems using available data.

During the analysis, it was found that the drinking water dataset does not include geographic identifiers such as latitude, longitude, or a shared location key. This limitation made it impossible to directly connect the two datasets.

Because of this, the project approach was adjusted. Instead of building a direct relationship between the datasets, the analysis was divided into two parts: one focusing on drinking water contaminant patterns and the other focusing on spatial and temporal patterns in surface water.

Data Sources and Description

This project uses two main datasets to analyze water quality patterns in Kansas: a drinking water dataset and a surface water dataset. These datasets were selected to support both statistical analysis and spatial analysis of contaminants.

Drinking Water Dataset: The drinking water dataset contains records from municipal water systems, including measurements of different contaminants. Each record represents a reported value for a specific contaminant, along with related information such as unit, regulatory limit, and collection date.

The key variables used in this dataset include:

- Contaminant name
- Reported concentration value (Highest Value)
- Measurement unit (ppm, mg/L, µg/L)
- Maximum Contaminant Level (MCL)
- Collection date and year
- Source or system identifier

One of the main challenges in this dataset was the inconsistency in units and value formats. Concentrations were reported using different units, which made direct comparison difficult. In addition, some values were recorded in non-numeric formats such as “<0.5” and “ND” (non-detect), which required cleaning before analysis.

After preprocessing, all concentration values were converted into a common unit (ppb), and only valid numeric records were retained. Duplicate entries were removed, and the dataset was prepared for further analysis.

Surface Water Dataset: The surface water dataset was obtained from the Water Quality Portal (WQP), which provides water quality measurements collected from monitoring locations across the United States.

The key variables used in this dataset include:

- Monitoring Location Identifier
- Monitoring Location Name
- Date of measurement
- Characteristic name (e.g., nitrate, pH, turbidity)
- Measurement value
- Unit
- Latitude and longitude

Unlike the drinking water dataset, this dataset includes geographic coordinates, which allows for spatial analysis. This makes it possible to visualize monitoring locations and analyze how contaminant levels vary across different areas.

After cleaning, the surface water dataset contained approximately 17,000 records across 164 monitoring locations. Records with missing geographic or measurement values were removed to ensure reliable analysis.

Focus on Nitrate: During initial analysis, nitrate was identified as one of the most important contaminants due to its significantly higher concentration levels compared to others. Because of this, nitrate was selected for detailed analysis across both datasets.

In the surface water dataset, nitrate records were filtered to create a subset of approximately 420 observations across 74 monitoring locations. This subset was used for spatial and temporal analysis.

Data Limitations: A key limitation identified in this project is the lack of geographic information in the drinking water dataset. Specifically, the dataset does not include latitude, longitude, or a shared identifier that can be used to link it with surface water monitoring locations.

Because of this, it is not possible to directly connect drinking water systems with specific surface water sources. This limitation affects the ability to perform combined spatial analysis across the two datasets.

Summary of Data Sources: In summary, the drinking water dataset provides detailed information on contaminant levels and trends within municipal systems, while the surface water dataset provides geographic context and supports spatial analysis. Although a direct linkage between the datasets is not possible, both datasets contribute to understanding water quality patterns in Kansas.

Data Preparation

The data preparation phase was essential for ensuring that both datasets were clean, consistent, and suitable for analysis. Since the datasets were collected from different sources, they contained inconsistencies in units, formats, and missing values that needed to be addressed before meaningful analysis could be performed.

Drinking Water Data Preparation: The drinking water dataset required significant preprocessing due to variations in how contaminant values were recorded. One of the main issues was the use of different measurement units such as ppm, mg/L, and µg/L. To make the data comparable across all contaminants, all values were converted into a common unit, parts per billion (ppb). This step ensured consistency and allowed accurate comparison of concentration levels.

Another challenge was handling non-numeric entries such as “<0.5” and “ND” (non-detect). These values were processed by extracting the numeric component where possible or

treating them as missing values. This helped prevent errors in calculations and ensured that only valid numeric values were used in analysis.

Duplicate records were removed to avoid repeated observations. In addition, rows with missing or invalid values in key fields such as contaminant name, concentration, and year were excluded. This improved the overall quality and reliability of the dataset.

To support analysis, new variables were created. The standardized concentration value (Highest_ppb) was used as the primary variable for analysis. Regulatory limits were also converted into ppb (MCL_ppb) to allow direct comparison with observed values. An exceedance indicator (Exceed_MCL) was created to identify cases where contaminant levels exceeded regulatory limits.

An outlier detection step was also performed using the interquartile range (IQR) method for each contaminant. This helped identify unusually high or low values, which are important for understanding variability in the data.

Surface Water Data Preparation:

The surface water dataset required a different type of preprocessing focused on preparing the data for spatial analysis. Relevant columns such as measurement value, characteristic name, date, latitude, and longitude were selected for analysis.

Measurement values and geographic coordinates were converted into numeric format to ensure consistency. Records with missing latitude or longitude values were removed, as they cannot be used for mapping or spatial analysis.

Date values were converted into a standard format, and a year variable was extracted to support trend analysis over time. The dataset included many different water quality characteristics, but nitrate was selected for detailed analysis because it showed importance in the drinking water dataset. A filtered subset of nitrate observations was created, which was used for spatial visualization and temporal analysis.

Summary: Overall, the data preparation phase ensured that both datasets were accurate, consistent, and ready for analysis. Standardizing units, handling missing values, and creating derived variables allowed for reliable comparisons and meaningful interpretation of results in later stages of the project.

Methodology

The methodology for this project was designed to analyze contaminant patterns in drinking water and surface water datasets using a combination of data preprocessing, exploratory analysis, and

basic statistical techniques. The approach was divided into two main parts: analysis of drinking water data and analysis of surface water data.

Drinking Water Analysis Approach:

The drinking water analysis began with exploratory data analysis (EDA) to understand the distribution and frequency of contaminants. This step helped identify the most reported contaminants, which were then selected for further analysis. Focusing on frequently occurring contaminants ensured that the results were based on sufficient data.

Trend analysis was performed to examine how contaminant concentrations change over time. Average concentration values were calculated for each year, and line plots were used to visualize trends. This helped identify whether certain contaminants showed increasing, decreasing, or stable patterns.

A simple linear regression model was also applied to evaluate whether time (year) could explain changes in contaminant concentration. This model was used as an initial attempt to identify overall trends in the dataset. However, since the dataset includes multiple contaminants with different behaviors and concentration ranges, the model was mainly used to test whether a general trend exists.

Additional analysis included evaluating regulatory exceedances by comparing observed values with Maximum Contaminant Levels (MCL). This helped identify contaminants that pose potential risk based on regulatory standards. Outlier analysis was also performed to detect extreme values and understand variability within the data.

Surface Water Analysis Approach:

The surface water analysis focused on spatial and temporal patterns using the available geographic information. Since the dataset includes latitude and longitude, monitoring locations were visualized using scatter plots to understand their distribution across Kansas.

Nitrate was selected for detailed analysis based on its importance in the drinking water dataset. A subset of nitrate observations was created and used for further analysis.

Spatial analysis was performed by plotting nitrate values across monitoring locations. This helped identify whether high concentrations were concentrated in specific regions or distributed across multiple locations.

Temporal analysis was conducted by calculating average nitrate values for each year and visualizing the trend using line plots. This provided insight into how nitrate levels change over time in surface water systems.

A site-level summary was also created to compare nitrate values across different monitoring locations. This allowed for a more detailed understanding of spatial variation.

Combined Approach:

Although both datasets were analyzed separately, the methodology was designed to maintain a consistent focus on key contaminants, particularly nitrate. The goal was to compare patterns observed in drinking water and surface water, even though a direct linkage between the datasets was not possible.

Overall, the methodology combines data cleaning, exploratory analysis, visualization, and simple modeling techniques to extract meaningful insights while ensuring that the analysis remains aligned with the available data.

Drinking Water Analysis

The drinking water analysis focuses on identifying key contaminants, examining their concentration levels, and understanding how these values change over time. This section presents the main findings based on exploratory data analysis and supporting visualizations.

Contaminant Frequency Analysis: The first step in the analysis was to identify the most frequently reported contaminants in the dataset. Understanding frequency is important because it helps determine which contaminants have sufficient data for meaningful analysis.

The results show that Barium, Fluoride, Selenium, Nitrate, and Chromium are the most reported contaminants. These contaminants appear significantly more often than others, making them suitable for further trend and concentration analysis.

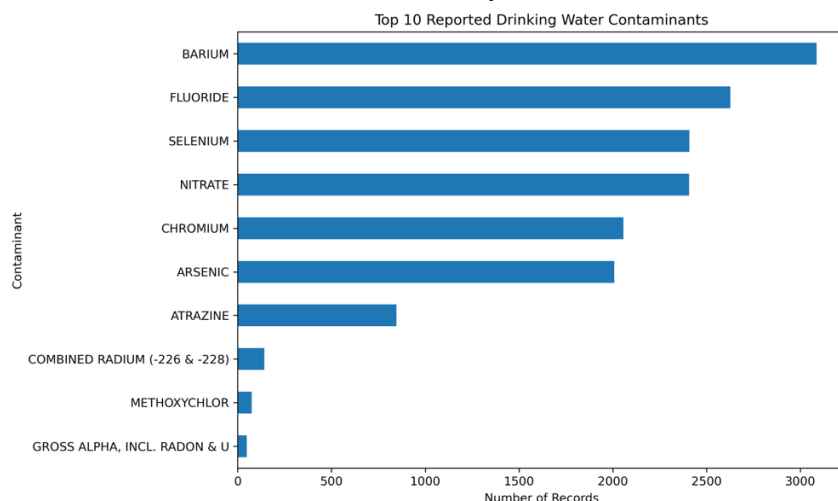


Figure 1: Top 10 Reported Drinking Water Contaminants

Trend Analysis of Top Contaminants: After identifying the most frequent contaminants, a trend analysis was performed to examine how their concentration levels change over time. Average concentration values were calculated for each year and plotted for comparison.

The results show that most contaminants, including Barium, Fluoride, Selenium, and Chromium, remain relatively stable over time with only minor variations. In contrast, nitrate shows a significantly different pattern, with much higher concentration levels compared to other contaminants.

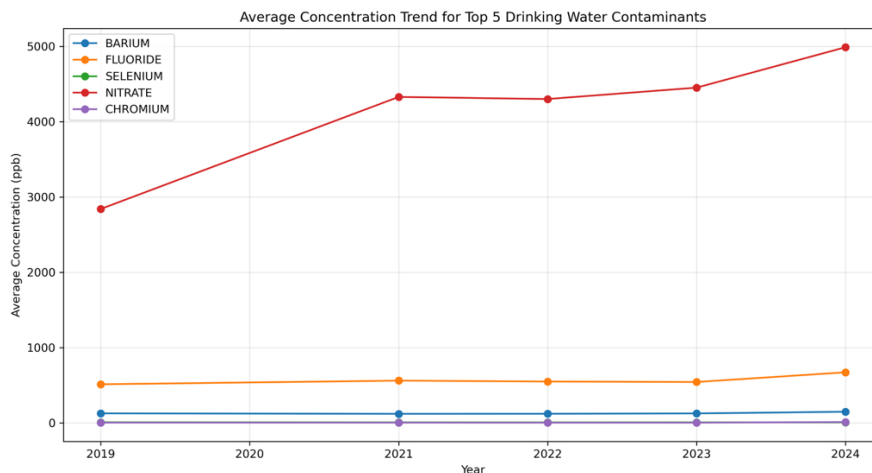


Figure 2: Average Concentration Trends for Top 5 Contaminants

Nitrate Analysis: Nitrate was analyzed separately because it showed the highest concentration levels among all contaminants. The trend analysis indicates that nitrate values increase from approximately 2800 ppb in 2019 to nearly 5000 ppb in 2024, with some fluctuations between years.

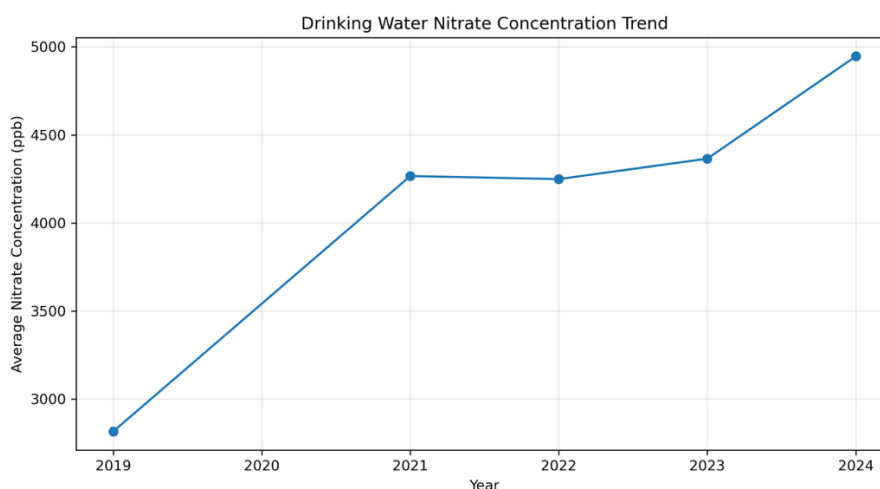


Figure 3: Drinking Water Nitrate Concentration Trend (2019–2024)

This pattern suggests that nitrate behaves differently from other contaminants and may require closer monitoring. The higher concentration levels also make nitrate a key contaminant of interest in this study.

Regression Analysis

A simple linear regression model was applied to evaluate whether year can explain variation in contaminant concentration when all contaminants are combined. The results show a very low R-squared value and a high p-value, indicating that there is no significant linear relationship.

This outcome suggests that combining all contaminants into a single model is not meaningful, as each contaminant has different behavior and concentration ranges. Therefore, contaminant-specific analysis provides more reliable insights.

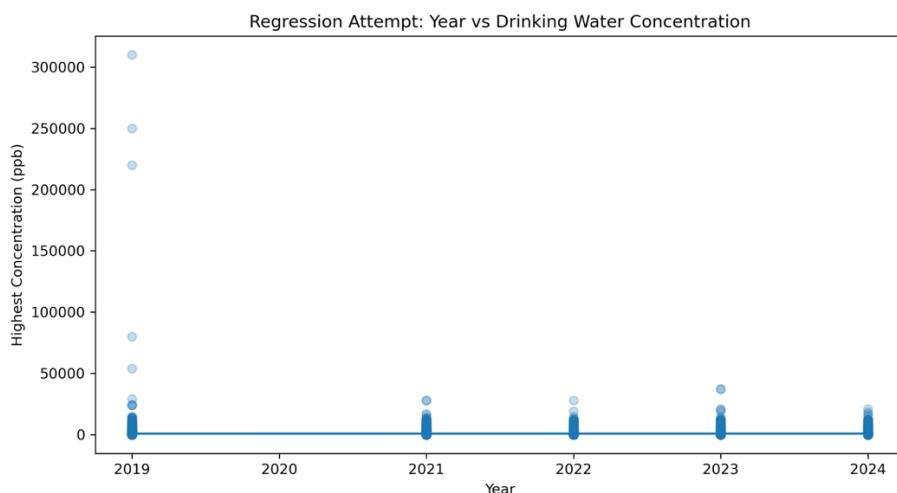


Figure 4: Regression Analysis (Year vs Concentration)

Exceedance Analysis

Exceedance analysis was conducted by comparing observed contaminant values with regulatory limits (MCL). This helps identify contaminants that may pose potential risks.

The results show that some contaminants have high exceedance rates but are based on a small number of observations. In contrast, nitrate has a moderate exceedance rate but a large number of records, making it more significant in terms of overall impact.

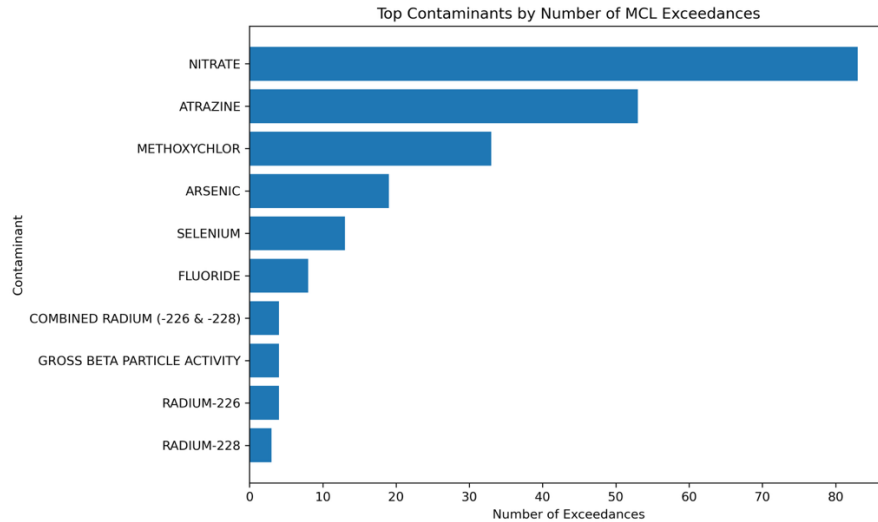


Figure 5: Contaminants by Number of MCL Exceedances

Summary of Drinking Water Analysis

Overall, the drinking water analysis shows that while several contaminants are commonly reported, nitrate stands out due to its higher concentration levels and noticeable variation over time. Most other contaminants remain relatively stable, indicating less variability.

The regression analysis confirms that aggregated modeling is not effective, and individual contaminant analysis is more meaningful. These findings highlight nitrate as a key contaminant that requires attention in further analysis.

Key findings from drinking water analysis:

- Nitrate shows significantly higher concentration levels compared to other contaminants.
- Most contaminants remain relatively stable over time.
- Aggregated regression does not provide meaningful insights.
- Exceedance patterns vary depending on both rate and number of observations.

Surface Water Analysis

The surface water analysis focuses on understanding how water quality measurements vary across different locations and over time. Unlike the drinking water dataset, this dataset includes geographic coordinates, which allows for spatial analysis of monitoring locations.

Monitoring Location Distribution

The first step in the surface water analysis was to visualize the distribution of monitoring locations. This helps understand how data is geographically spread across the state and whether certain areas have more coverage than others.

The plot shows that monitoring locations are distributed across Kansas, although the distribution is not uniform. Some regions have a higher concentration of monitoring sites, while others have fewer. This variation in coverage should be considered when interpreting spatial results.

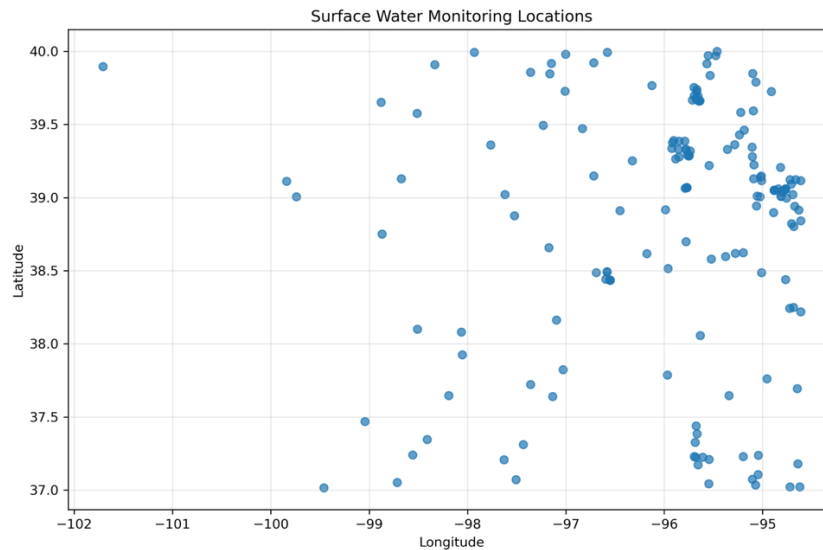


Figure 6: Distribution of Surface Water Monitoring Locations

Surface Water Characteristics:

The dataset includes a wide range of measured characteristics such as turbidity, pH, dissolved oxygen, temperature, and nitrate. These variables provide a general view of water quality conditions across monitoring sites.

Understanding the most frequently measured characteristics helps identify which variables are suitable for detailed analysis. Among these, nitrate was selected for further investigation because of its importance in the drinking water analysis.

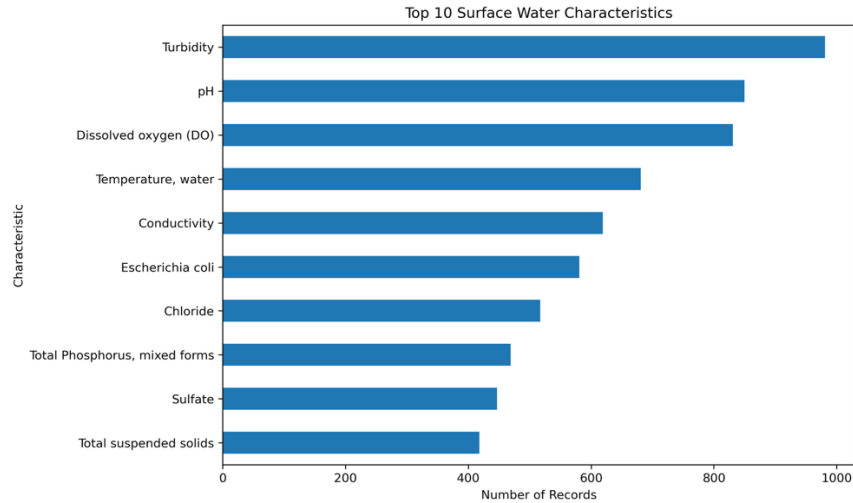


Figure 7: Most Frequently Measured Surface Water Characteristics

Nitrate Spatial Analysis

A subset of nitrate observations was created to analyze its spatial distribution. The dataset contains approximately 420 nitrate measurements across 74 monitoring locations. The spatial plot shows that nitrate values vary across different locations. Some sites have higher concentration values, while others remain lower. However, there is no single region where high nitrate values are consistently concentrated. This indicates that nitrate contamination is distributed across multiple locations rather than being limited to a specific hotspot.

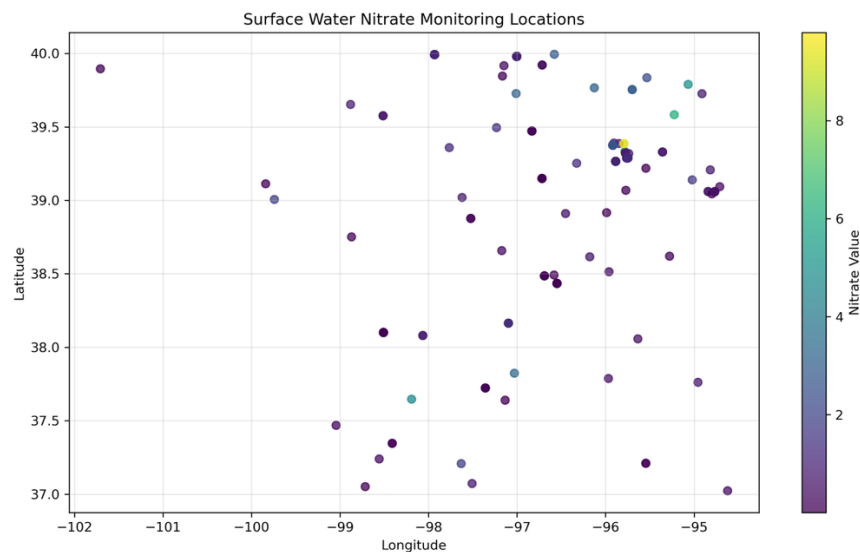


Figure 8: Spatial Distribution of Nitrate in Surface Water

Nitrate Temporal Analysis

To understand how nitrate levels change over time, a yearly trend analysis was performed using average values. The results show that nitrate concentrations fluctuate between approximately 0.8 and 1.7 across different years.

There is no clear increasing or decreasing trend, and the variation suggests that nitrate levels in surface water are influenced by changing environmental conditions rather than a consistent pattern over time.

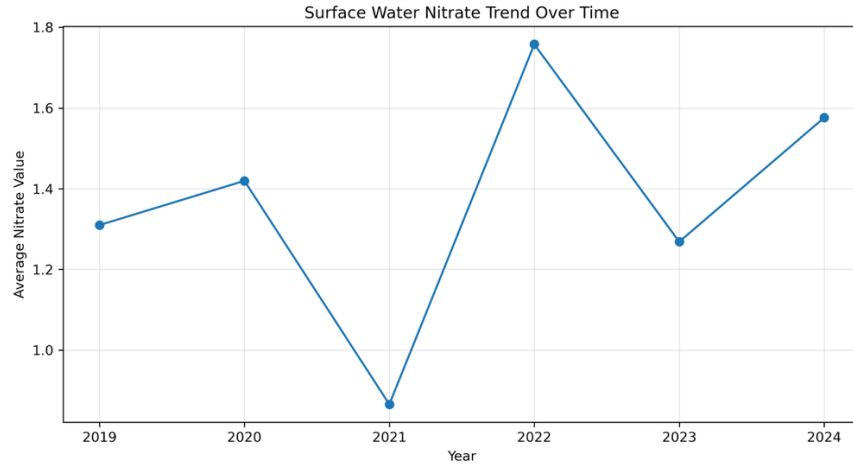


Figure 9: Surface Water Nitrate Trend Over Time

Site-Level Nitrate Analysis

A site-level summary was created to examine how nitrate values differ across monitoring locations. This analysis helps identify whether certain sites consistently show higher concentration levels.

The results indicate that some locations have higher average nitrate values than others. However, these higher values are not concentrated in a specific region, which supports the earlier observation that nitrate is distributed across multiple locations.

MonitoringLocationID	Records	Mean Value	Median Value	Max. Value	Latitude	Longitude
PRAIRIEBAND-MW001	2	6.7	6.7	6.7	39.3888	-95.7933
PRAIRIEBAND-MW008	2	6.7	6.7	6.7	39.3318	-95.7814
NARS.WQX-NRS.KS-10435	1	6.19	6.19	6.19	39.58314258	-95.25446835
NARS.WQX-NRS.KS-10449	1	5.02	5.02	5.02	39.78985916	-95.06722568
NARS.WQX-NRS.KS-10249	1	3.63	3.63	3.63	37.64535546	-98.19165595
NARS.WQX-NRS.KS-10248	1	3.3074	3.3074	3.3074	37.62311746	-97.03033315
NARS.WQX-NRS.KS-10210	2	3.1957	3.1957	3.6303	39.76447276	-98.68803365
NARS.WQX-NRS.KS-10369	1	3.1	3.1	3.1	38.72738834	-97.01089354
NARS.WQX-NRS.KS-10404	1	3.01	3.01	3.01	39.76648562	-98.12754016
NARS.WQX-NRS.KS-10205	1	2.7479	2.7479	2.7479	39.59441529	-96.57905059
NARS.WQX-NRS.KS-10443	1	2.26	2.26	2.26	39.83518263	-98.53311662
NARS.WQX-NRS.KS-10469	1	1.84	1.84	1.84	39.00531846	-99.74227905
NARS.WQX-NWC.KS-10139	1	1.63	1.63	1.63	37.20753864	-97.62111656
21NEB001.WQX-SLB11BLUE000	155	1.63989677	1.73	7.55	39.580149	-97.003893
NARS.WQX-NRS.KS-10139	1	1.63	1.63	1.63	39.139578	-98.020856
21NEB001.WQX-SRE1REPUB000	87	1.486218391	1.25	3.74	39.562569	-97.931967
NARS.WQX-NRS.KS-10050	1	1.3473	1.3473	1.3473	39.49522569	-97.23139218
NARS.WQX-NRS.KS-10046	1	1.27	1.27	1.27	39.28252179	-96.32659742
NARS.WQX-NRS.KS-10360	1	1.27	1.27	1.27	39.35267743	-97.76558672
PRAIRIEBAND-SC07	38	1.151111111	1.05	3.3	39.2658	-95.8864
PRAIRIEBAND-SC05	25	1.1228	0.83	2.9	39.2758	-95.9183
PRAIRIEBAND-LSC06	10	1.107	1.15	1.8	39.2888	-95.7483
PRAIRIEBAND-MW10	1	1.1	1.1	1.1	39.3873	-95.8494
NARS.WQX-NRS.KS-10269	2	1.03485	1.03485	2.0217	39.52164243	-96.71631982
PRAIRIEBAND-MW009	1	0.97	0.97	0.97	39.3904	-95.9053
NARS.WQX-NRS.KS-10255	1	0.9048	0.9048	0.9048	39.65317317	-98.88144118
NARS.WQX-NRS.KS-10226	2	0.8935	0.8935	1.1667	39.32949078	-95.35088326
NARS.WQX-NRS.KS-10208	1	0.8141	0.8141	0.8141	39.01891425	-97.62109054
NARS.WQX-NRS.KS-10222	1	0.8079	0.8079	0.8079	39.20750093	-94.81716042
NARS.WQX-NRS.KS-10039	1	0.7985	0.7985	0.7985	39.2824414	-94.91122093
NARS.WQX-NRS.KS-10431	1	0.724	0.724	0.724	38.61579999	-96.17843863
NARS.WQX-NRS.KS-10246	1	0.7193	0.7193	0.7193	37.76116475	-94.5652043
NARS.WQX-NRS.KS-10199	1	0.6977	0.6977	0.6977	37.07185357	-97.50849573
EPA.R7.WQX-RSTIP09	2	0.655666667	0.27	1.55	38.163668	-97.057927
NARS.WQX-NRS.KS-10051	2	0.6841	0.7043	0.96027253	-94.84194972	
PRAIRIEBAND-LSC03	8	0.6675	1.2	39.3225	-98.7758	
NARS.WQX-NRS.KS-10196	2	0.6445	0.6445	0.74	39.57629487	-98.51444655
PRAIRIEBAND-MW003	4	0.61	0.61	0.61	39.318614	-95.738642
PRAIRIEBAND-LSC04	4	0.57	0.6	0.63	39.2887	-95.7617
NARS.WQX-NRS.KS-10264	1	0.5171	0.5171	0.5171	38.91041431	-96.45214172
NARS.WQX-NWC.KS-10102	2	0.5151	0.5151	0.991	38.07983	-98.06508
NARS.WQX-NRS.KS-10250	1	0.5087	0.5087	0.5087	39.05708386	-95.63541796
EPA.R7.WQX-KVC05	1	0.491	0.491	0.491	39.05990628	-95.6064306
NARS.WQX-NWC.KS-10143	1	0.478	0.478	0.478	37.7872377	-98.987677
NARS.WQX-NRS.KS-10293	1	0.4029	0.4029	0.4029	38.51416207	-95.98179912
EPA.R7.WQX-C012	1	0.374	0.374	0.374	39.083076	-94.71003265
NARS.WQX-NRS.KS-10289	1	0.3634	0.3634	0.3634	37.02237885	-94.6211974
EPA.R7.WQX-RSTIP0	4	0.3576	0.3576	0.3576	37.236466	-98.358466
NARS.WQX-NRS.KS-10252	1	0.332	0.332	0.332	38.65767114	-97.17331128
NARS.WQX-NRS.KS-10129	1	0.331	0.331	0.331	37.253256	-98.557531
NARS.WQX-NRS.KS-10361	1	0.33	0.33	0.33	39.51711927	-97.14841713
EPA.R7.WQX-RSTIP03	5	0.2962	0.132	0.768	38.475247	-98.7185223
NARS.WQX-NRS.KS-10448	1	0.215	0.215	0.215	37.4680573	-99.04453627
EPA.R7.WQX-KVC04	1	0.195	0.195	0.195	39.05770136	-94.76787234
NARS.WQX-NRS.KS-10245	1	0.189	0.189	0.189	38.75125952	-98.87007228
NARS.WQX-NRS.KS-10262	1	0.1881	0.1881	0.1881	36.61585251	-95.27723261
NARS.WQX-NRS.KS-10093	2	0.1469	0.1469	0.1822	37.20970471	-95.54539085
EPA.R7.WQX-RSTIP04	2	0.1435	0.1435	0.1435	38.4683076	-97.523127
EPA.R7.WQX-KVC06	1	0.139	0.139	0.139	39.060303	-94.7665188
EPA.R7.WQX-RSTIP08	5	0.0955	0.131	0.262	39.0444982	-96.691022
NARS.WQX-NRS.KS-10446	1	0.079	0.079	0.079	39.84655733	-97.16511325
EPA.R7.WQX-RSTIP02	2	0.0752	0.0752	0.0752	39.0444982	-94.7665188
EPA.R7.WQX-RSTIP05	10	0.065	0.02	0.147	38.10035	-98.509398
EPA.R7.WQX-RSTIP07	5	0.0605	0.02	0.22	38.145088	-98.718446
NARS.WQX-NRS.KS-10280	1	0.0591	0.02	0.219	38.43461	-98.54928
NARS.WQX-NRS.KS-10236	1	0.0548	0.0548	0.0548	39.05770136	-98.7185223
EPA.R7.WQX-B02	1	0.051	0.051	0.051	39.89576845	-101.7077863
NARS.WQX-NRS.KS-10412	1	0.0432	0.0432	0.0432	38.6585004	-98.77104832
NARS.WQX-NRS.KS-10412	1	0.03	0.03	0.03	38.91834574	-95.9883252
NARS.WQX-NRS.KS-10484	1	0.01	0.01	0.01	39.0887246	-95.84474822
NARS.WQX-NRS.KS-10098	1	0.008	0.008	0.008	38.49167	-98.58222
NARS.WQX-NRS.KS-10271	2	0.0072	0.0072	0.0072	38.10035	-98.51008
NARS.WQX-NRS.KS-10271	1	0.0054	0.0054	0.0054	39.11269578	-99.84059319
NARS.WQX-NWC.KS-10142	1	0.0016	0.0016	0.0016	37.639357	-97.134782

Table 1: Summary of Nitrate Values by Monitoring Location

Overall, the surface water analysis shows that monitoring locations are spread across Kansas and provide sufficient coverage for spatial analysis. Nitrate values vary across locations and fluctuate over time, with no clear linear trend.

The spatial and site-level analysis indicates that nitrate contamination is not limited to a single hotspot but is distributed across multiple monitoring locations. These findings provide useful context for understanding contaminant behavior in surface water systems.

Key findings from surface water analysis:

- Monitoring locations are distributed across Kansas with uneven coverage.
- Nitrate concentrations vary across locations with no single hotspot.
- Temporal trends show fluctuations rather than a consistent pattern.
- Site-level analysis confirms variation across multiple locations.

Comparative Discussion

This section compares the key findings from the drinking water and surface water analyses to understand how contaminant patterns differ and where they show similarities. Although a direct linkage between the datasets is not possible, comparing the results provides useful insights into overall water quality behavior.

From the drinking water analysis, nitrate was identified as the most significant contaminant based on its higher concentration levels and noticeable variation over time. The trend showed an increase from approximately 2800 ppb in 2019 to nearly 5000 ppb in 2024. In contrast, most other contaminants such as Barium, Fluoride, Selenium, and Chromium remained relatively stable over time.

In the surface water analysis, nitrate was also examined as the primary contaminant of interest. However, the behavior observed in surface water was different. Instead of showing a clear increasing trend, nitrate values fluctuated between approximately 0.8 and 1.7 across different years. This indicates that surface water nitrate levels vary over time but do not follow a consistent pattern.

Spatial analysis further highlighted differences in how nitrate appears in surface water compared to drinking water. In surface water, nitrate concentrations were distributed across multiple monitoring locations, with no single region consistently showing the highest values. This

suggests that contamination is spread across different areas rather than being concentrated in a specific hotspot.

While both datasets highlight nitrate as an important contaminant, the patterns observed in each system are not directly comparable due to differences in data structure, measurement context, and lack of shared identifiers. The drinking water dataset reflects treated water supplied to communities, while the surface water dataset represents raw environmental measurements.

A key limitation in this comparison is the absence of geographic information in the drinking water dataset. Without latitude, longitude, or a shared location identifier, it is not possible to match specific drinking water systems with nearby surface water monitoring sites. Because of this, the analysis cannot establish a direct relationship between contamination in surface water and drinking water quality.

Despite this limitation, comparing the two analyses provides useful context. The presence of nitrate in both datasets suggests that it is a widespread contaminant of concern. However, the differences in trends and spatial patterns indicate that additional data would be required to fully understand how surface water contamination influences drinking water systems.

Overall, this comparison highlights the importance of data availability and structure when attempting to analyze interconnected environmental systems. It also demonstrates that careful interpretation is necessary to avoid drawing unsupported conclusions.

Limitations

This project provides useful insights into water quality patterns; however, several limitations must be considered:

- The drinking water dataset does not include geographic identifiers such as latitude or longitude, making it impossible to directly link it with surface water monitoring locations.
- The drinking water and surface water datasets represent different contexts, where one reflects treated water and the other represents raw environmental measurements, making direct comparison challenging.
- Some records contained missing values or non-standard formats, which required cleaning and removal of certain observations.
- The analysis primarily focused on a limited set of contaminants, with detailed emphasis on nitrate, rather than covering all possible contaminants.

- External environmental factors such as seasonal variation, rainfall, and land use were not included, which may influence contaminant levels.

Despite these limitations, the analysis provides meaningful insights based on the available data.

Future Work and Recommendations

Based on the findings of this project, the following improvements can be considered for future work:

- Include geographic identifiers for drinking water systems to enable direct linkage with surface water data.
- Incorporate additional environmental variables such as rainfall, land use, and seasonal factors to better explain variations in contaminant levels.
- Expand the analysis to include a broader set of contaminants beyond nitrate.
- Apply more advanced modeling techniques if more detailed and connected data becomes available.
- Improve data consistency and standardization across sources to support integrated analysis.
- These improvements can help build a more complete understanding of water quality systems in future studies.

Conclusion

This project analyzed water quality data in Kansas to understand patterns in drinking water contaminants and to examine how these patterns relate to surface water conditions. Using both statistical and spatial analysis approaches, the study provides insights into contaminant behavior across two different types of water systems.

The drinking water analysis identified several commonly reported contaminants, including Barium, Fluoride, Selenium, Nitrate, and Chromium. Among these, nitrate stands out due to its significantly higher concentration levels and noticeable variation over time. The trend shows an increase from approximately 2800 ppb in 2019 to nearly 5000 ppb in 2024, indicating that nitrate is an important contaminant that may require closer monitoring.

The surface water analysis provided additional context by examining spatial and temporal patterns using geographic data. Monitoring locations were distributed across Kansas, and nitrate concentrations were found to vary across sites and fluctuate over time. Unlike the drinking water dataset, the surface water data did not show a consistent increasing trend, but rather a pattern of variation influenced by changing conditions.

Although both datasets highlight nitrate as a key contaminant, a direct relationship between surface water and drinking water systems could not be established due to the absence of geographic identifiers in the drinking water dataset. This limitation prevents direct spatial linkage and emphasizes the importance of data availability in environmental analysis.

Overall, the project demonstrates the importance of data preprocessing, exploratory analysis, and visualization in understanding environmental datasets. It also highlights how real-world data limitations can influence the scope of analysis and require adjustments in methodology.

The findings from this study provide a useful foundation for future research aimed at integrating multiple data sources and developing a more complete understanding of water quality systems.

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